NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

Pathology of Lung and Pleural Neoplasms

PPOM2

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Heal.
Innovate.
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Session Objectives

- 1. Describe and discuss the **general features and classifications of lung carcinoma** and evaluate its relationship to smoking
- 2. Discuss and compare the pathological features of **adenocarcinoma, squamous cell carcinoma, small cell carcinoma, and large cell carcinoma and correlate and describe their precursor lesions**
- 3. Discuss, describe, and distinguish some **genetic** aspects of the various types of **lung carcinomas**
- 4. Describe and explain the pathological features of **carcinoid tumor of the lung and metastatic nodules**
- 5. Discuss and describe the clinicopathological aspects, microscopic, and gross features of pleural neoplasms, **specifically malignant mesothelioma**

Topics

- General info/genetics
- Precursor lesions
- Adenocarcinoma (Non-small cell)
- Squamous cell carcinoma (Non-small cell)
- Small Cell carcinoma
- Large cell carcinoma
- Neuroendocrine tumors (Carcinoids)
- Metastatic tumors to lung
- Hamartoma
- Malignant mesothelioma

Lung Carcinoma

- Small Cell Carcinoma (15%)
- Non-small Cell Cancer [NSCLC]
  - –Squamous-cell carcinoma (20%)
  - –Adenocarcinoma (50%)
  - –Large-cell carcinoma (2%)
  - –Other/mixed (13%)

Features visible on light microscopy

Usually sufficient for most clinical needs

Source: Robbins and Cotran, Pathologic Basis of Disease, 11th ed, 2025, Ch. 15

Carcinoma of the Lung

- Most lung tumors (90 to 95%) are carcinomas
- Most frequently diagnosed major cancer in the world and most common cause of cancer mortality
- Older adults, 55 to 84 years old
- Major cause – cigarette smoking; other etiologies include radiation, asbestos (lung cancer and mesothelioma), radon
- 25% of lung cancers world wide are in non-smokers – tend to be women, tend to be adenocarcinomas – usually EGFR mutations
- Metastases common to liver (30 to 50%), brain (20%), bone (20%), and **adrenals (>50%)**

Source: Robbins and Cotran, Pathologic Basis of Disease, 11th ed, 2025, Ch. 15

Carcinoma of the Lung- Genetics

- Squamous cell and small cell especially associated with smoking

- Squamous cell shows **highest frequency of p53 mutations of all histological types, also**
  - CDKN2 (tumor suppressor gene) mutated in 65%
  - amplification of FGFR1

- **Small cell** also has frequent loss-of-function mutations involving **p53; RB loss-of-function; amplification of genes in MYC family**

- **Adenocarcinomas** may have gain-of-function mutations in multiple genes encoding receptor tyrosine kinases including **EGFR, ALK, ROS1, MET, RET, also:**
  - gain-of-function mutations in serine/threonine kinases such as BRAF, PIK3
  - KRAS gain-of-function mutations (30% of tumors)

Source: Robbins and Cotran, Pathologic Basis of Disease, 11th ed, 2025, Ch. 15

WHY ARE GENETICS IMPORTANT?

- PROGNOSIS AND TREATMENT
- Patients with adenocarcinoma and activating EGFR mutations (15%) treated with specific kinase inhibitors which prolong survival
- However, KRAS mutations (30% of adenocarcinomas) usually have worse prognosis – although treatments on horizon

Robbins and Cotran, Pathologic Basis of Disease, 11th ed. 2025

Precursor lesions

- Atypical adenomatous hyperplasia
- Adenocarcinom-in-situ
- Squamous dysplasia/CIS
- Diffuse idiopathic pulmonary neuroendocrine cell hyperplasia

These are preinvasive and do not mean progression to cancer is inevitable

Source: Robbins and Cotran, Pathologic Basis of Disease, 11th ed, 2025, Ch. 15

Case question

A 53 year old woman with no history of smoking presents with a 3 month history of chest pain, cough, and mild fever. A chest Xray reveals a peripheral mass in the left upper lobe. When the mass is resected, the cut surface is mucinous and histological examination reveals well formed glands. What is the most likely diagnosis?

A. large cell carcinoma

B. mesothelioma

C. small cell carcinoma

D. adenocarcinoma

E. squamous cell carcinoma

Adenocarcinoma

usually **peripheral location**



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http://library.med.utah.edu/WebPath/LUNGHTML/LUNG070.html



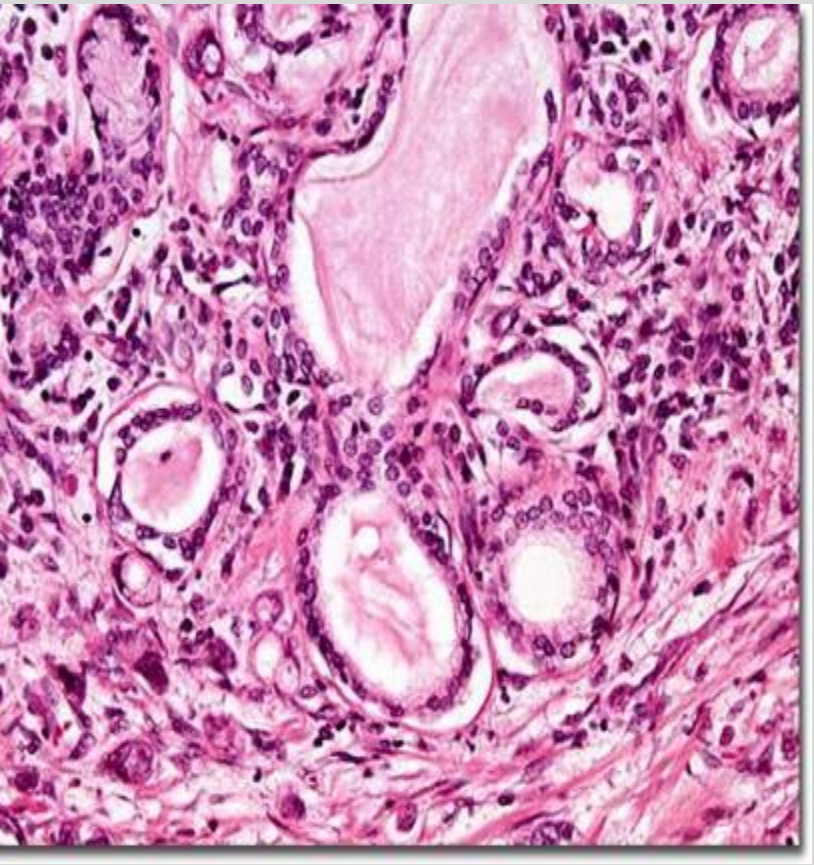
https://lh3.googleusercontent.com/z3MWGsBAd9jHgqNEbty1vkvZbEenQG_Xsd6yppqKrE7QfjNzmdWfQiLUYdtM3f6tSo8O=s127

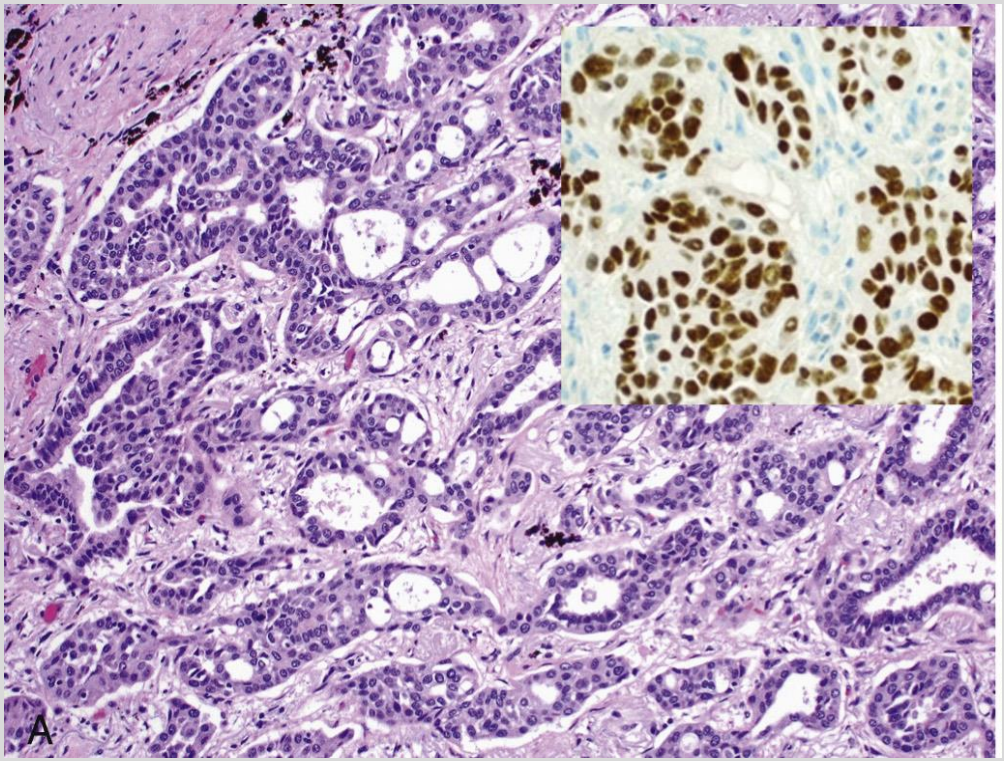
Lung Adenocarcinoma

- Most common type in women and non-smokers
- Glandular differentiation/mucin production
- Grows more slowly than squamous cell Ca but tends to metastasize widely and earlier
- **KRAS mutations seen, esp. if smoker, correlate with worse outcome**
- **EGFR mutations and amplications (implication for inhibitor treatment)**

Source: Robbins and Cotran, Pathologic Basis of Disease, 11th ed, 2025, Ch. 15

Lung adenocarcinomas show glandular differentiation microscopically and about 80% have mucin. There can be well differentiated glands or subtypes with papillary structures or solid areas; **+TTF-1**





https://www.microscopyu.com/staticgallery/pathology/adenocarcinomaoflung20x03.html

Source: Robbins and Cotran, Pathologic Basis of Disease, 11th ed, 2025, Ch. 15

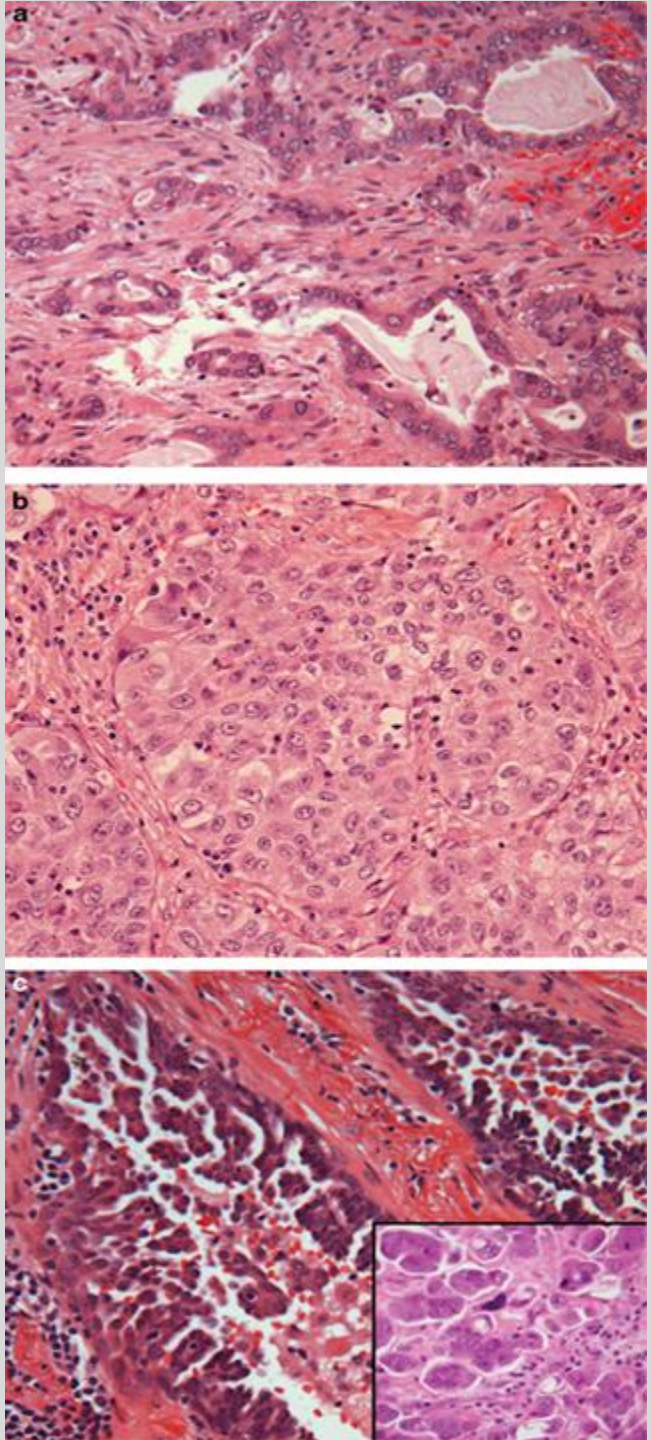
WHO Classification Invasive Adenocarcinoma

- Lepidic predominant (formerly non-mucinous BAC pattern)
- Acinar predominant
- Papillary predominant
- Micropapillary predominant
- Solid predominant

- Histological subtype is associated with frequency of EGFR mutation but does not replace mutation testing

Common histologies of invasive lung **adenocarcinoma.**

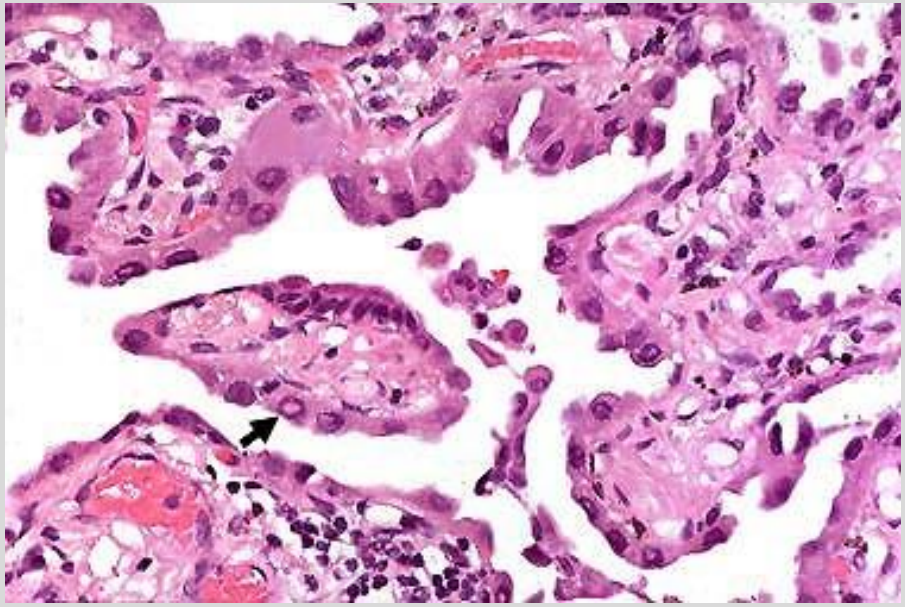
- The irregular gland formations of acinar pattern.
- The confluent solid growing nests of cells of solid pattern.
- Finger-like tufted projections of micropapillary pattern with the more obviously invasive micropapillary adenocarcinoma in the inset (H&E stain)



http://www.nature.com/modpathol/journal/v25/n1s/fig_tab/modpathol2011151f6.html

Atypical Adenomatous Hyperplasia

- **Small (<5 mm) precursor lesion**
- Dysplastic pneumocytes lining mildly fibrotic alveolar walls
- Single or multiple

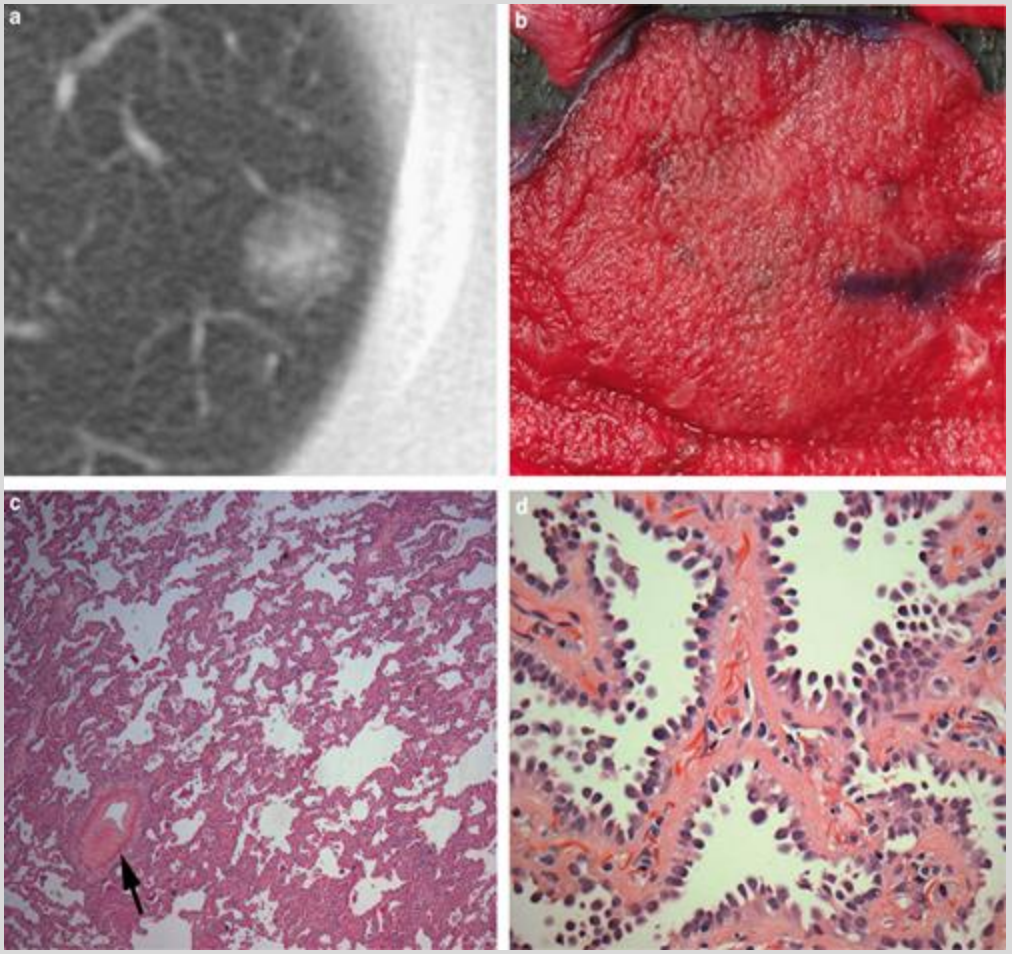


Epithelial cell nuclei show random orientation in the cytoplasm, slight hyperchromaticity, and irregularity in shape.

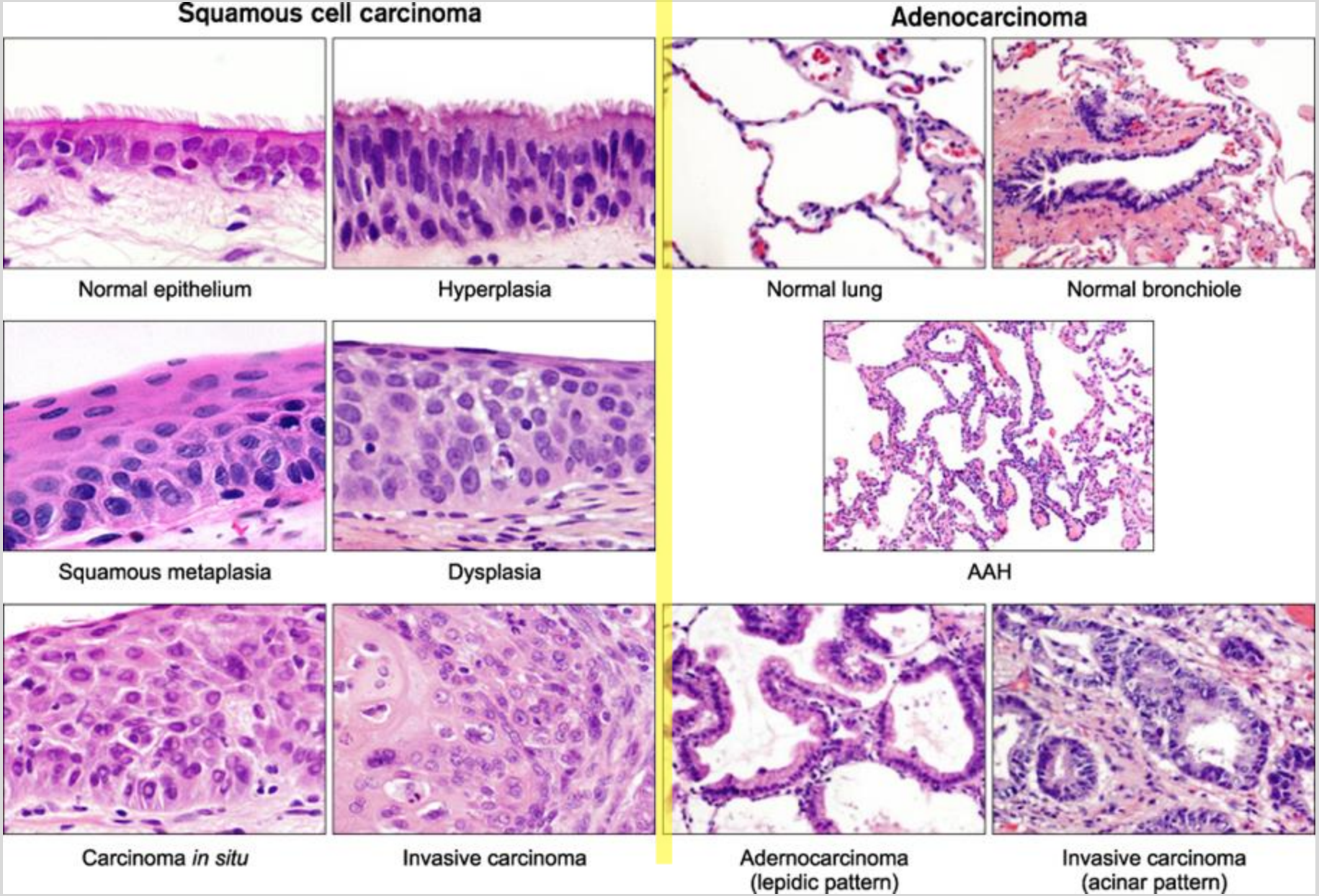
http://pathhsw5m54.ucsf.edu/case24/hyperplasia.html

Adenocarcinoma-in-situ

- Formerly bronchioalveolar carcinoma
- < 3cm
- Composed entirely of dysplastic cells growing along preexisting alveolar septae ("lepidic": Greek lepis - scale)
- Mucinous or nonmucinous types
- *Tumors < 3cm, with invasive component < 5mm, associated with scarring and peripheral lepidic growth pattern are called microinvasive adenocarcinoma – better prognosis*

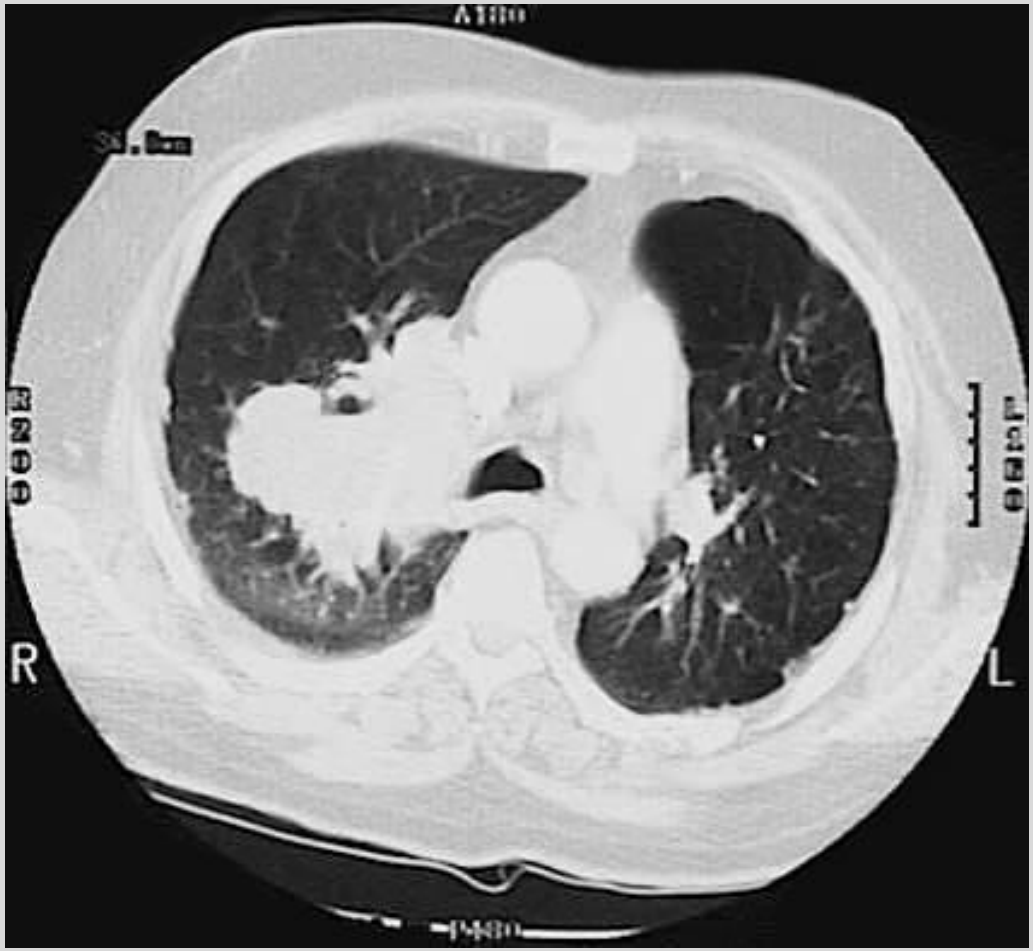


http://www.nature.com/modpathol/journal/v25/n1s/full/modpathol2011151a.html



http://synapse.koreamed.org/ViewImage.php?Type=F&aid=376077&id=F2&afn=160_JLC_11_1_12&fn=jlc-11-12-g002_0160JLC

Large lung mass involving mediastinum (central location)



https://www.studyblue.com/notes/note/n/pathology-3-pulmonary-2/deck/12967452

Squamous cell carcinoma of Lung –

most common in men, close correlation with smoking



Squamous cell carcinoma of the lung arising centrally in the lung obstructing right main bronchus - very firm with pale white to tan cut surface.

(WebPath)

Some squamous cell carcinomas may have necrotic foci and show areas of cavitation

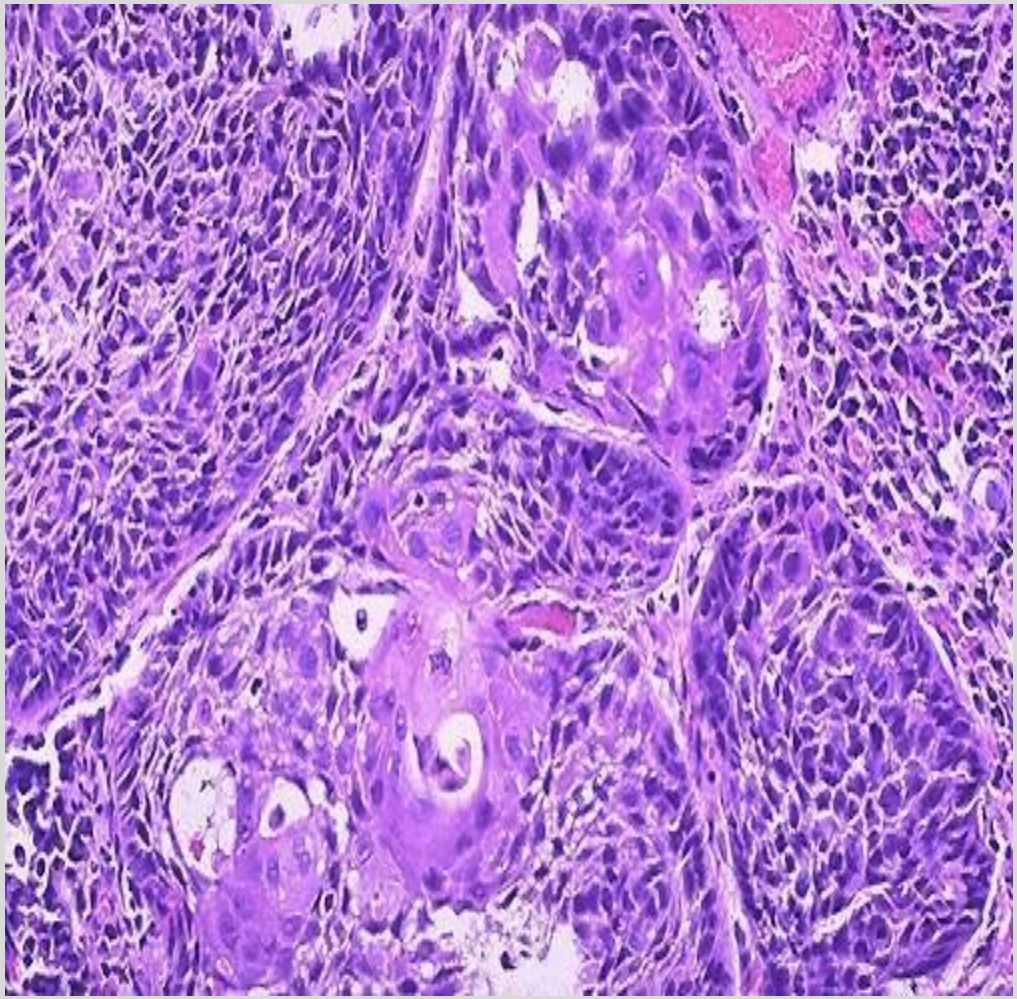
http://library.med.utah.edu/WebPath/LUNGHTML/LUNG067.html

Squamous cell CA of lung

- Most commonly found in men and closely correlated with smoking
- Histo shows **keratinization and intracellular bridges, esp. if well-differentiated (squamous pearls)**
- Usually **arises centrally**
- **Most have p53 mutations**

Source: Robbins and Cotran, Pathologic Basis of Disease, 11th ed, 2025, Ch. 15

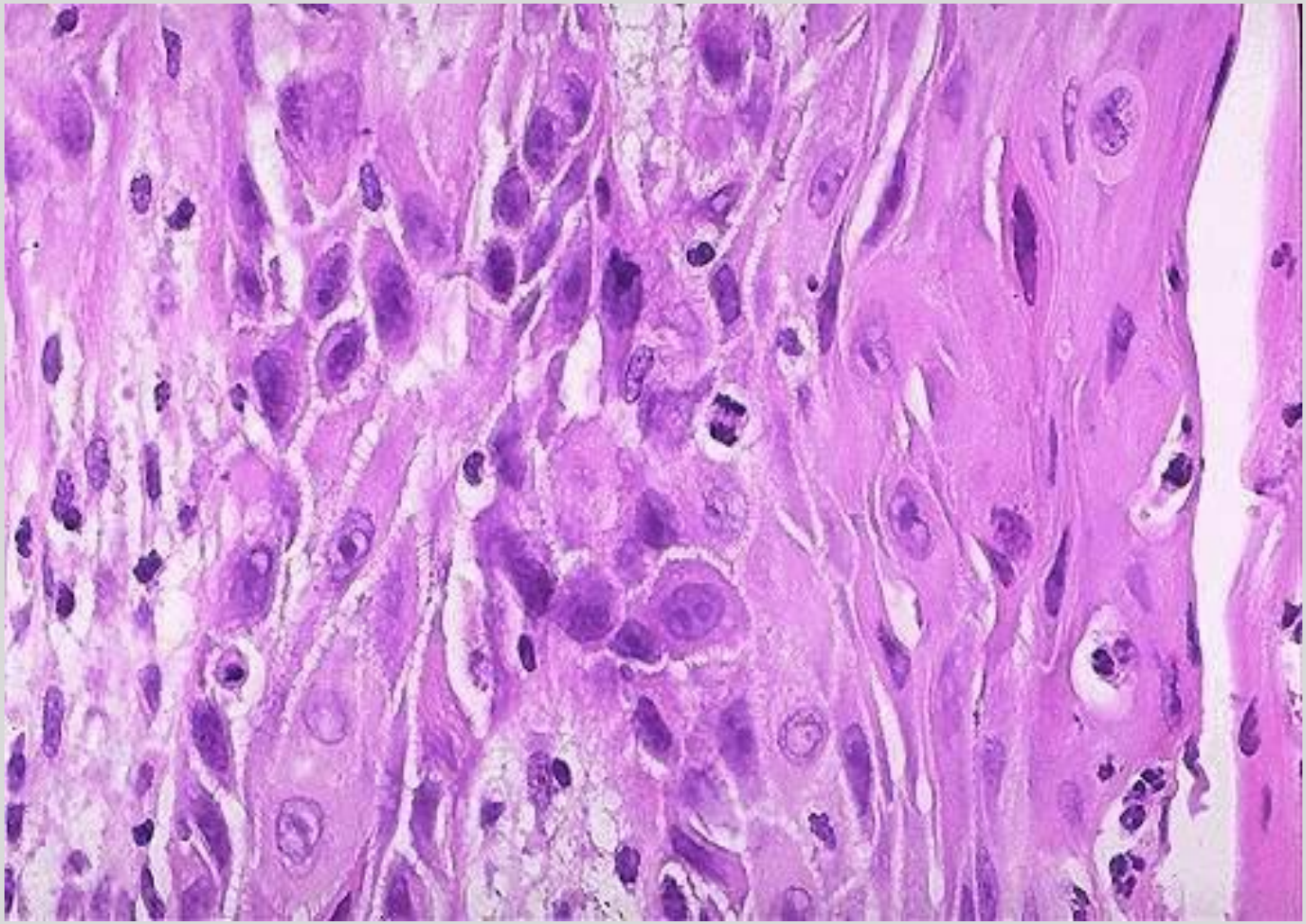
Squamous cell carcinoma



Presence of keratinization (squamous pearls) and/or intercellular bridges, especially in well-differentiated tumors

http://www.slideshare.net/csbrprasad/carcinoma-lung-15952310

Associated with hypercalcemia due to elaboration of PTH-related peptide



http://www.slideshare.net/csbrprasad/carcinoma-lung-15952310

Case question

A 52 year old woman has a one year history of truncal obesity and moderate depression. PE shows hirsutism and moon facies. Endocrinologic studies reveal hypokalemia, high plasma corticotropin levels, and increased serum and urine cortisol. CT scan of the thorax demonstrates a hilar mass. A transbronchial lung biopsy is performed (see next slide). EM shows neuroendocrine granules within the cytoplasm of some tumor cells. What is the most likely diagnosis?

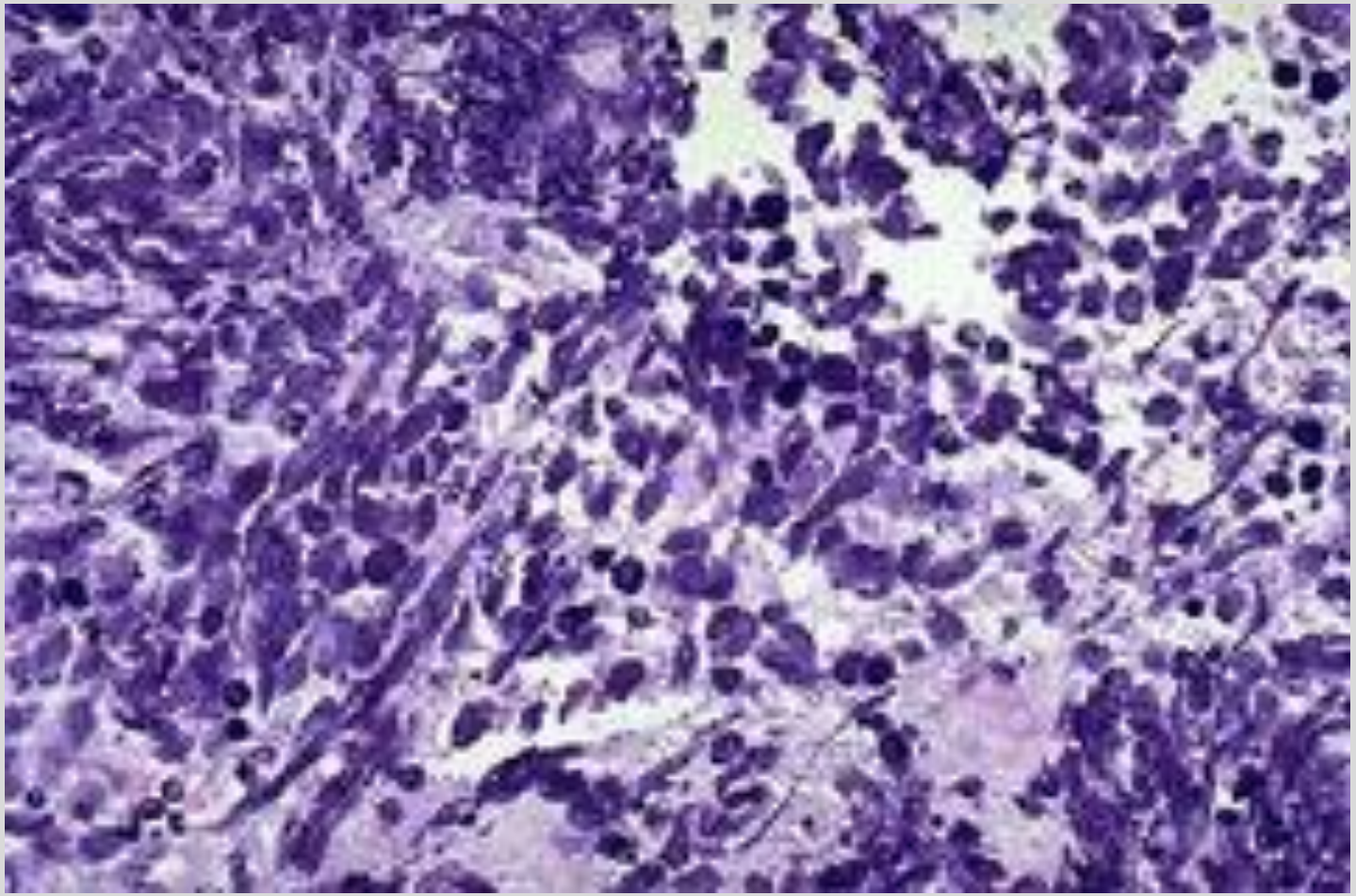
A. Large cell carcinoma

B. Adenocarcinoma

C. Squamous cell carcinoma

D. Small cell carcinoma

E. Metastatic carcinoma



AZZOPARDI EFFECT – Basophilic staining of vascular walls due to encrustation by DNA from necrotic tumor cells

http://www.slideshare.net/csbrprasad/carcinoma-lung-15952310

Small Cell Carcinoma of Lung



- Soft, lobulated, white to tan appearance.
- Small cell carcinomas are very aggressive, highly malignant, and often metastasize widely before the primary tumor mass in the lung reaches a large size. (WebPath)
- p53 and RB tumor suppressor gene mutations

http://missinglink.ucsf.edu/lm/ids_103_lung_cancer/subpages/footer_pages/ILM_printerfriendly.pdf

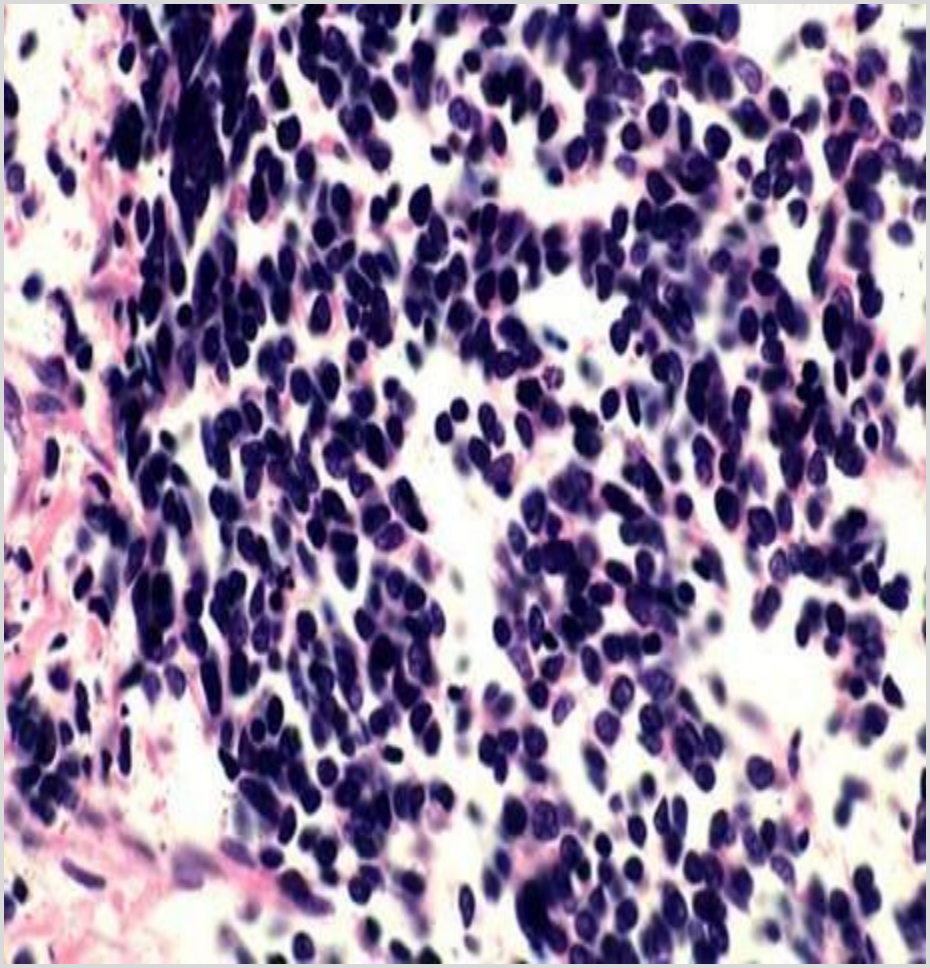
Small Cell Carcinoma

- Highly malignant, aggressive, 5-year survival rate very low
- **Neuroendocrine:**
  - histo shows small cells with scant cytoplasm, ill-defined cell borders
  - finely granular nuclear chromatin (“salt and pepper”)
  - absent/inconspicuous nucleoli
  - cells round to oval with nuclear molding
  - high mitotic rate
  - Azzopardi effect (basophilic staining of vascular walls due to encrustation by DNA from necrotic tumor cells)
- Strongest relationship to smoking
- **Dense-core neurosecretory granules (may be seen by EM) can secrete hormones ectopically;**

  + chromogranin,+synaptophysin
- **P53 and RB1 tumor suppressor genes frequently mutated**; high levels of BCL-2

Source: Robbins and Cotran, Pathologic Basis of Disease, 11th ed, 2025, Ch. 15

Small Cell carcinoma



- Small, blue cells with scant cytoplasm, absent or inconspicuous nucleoli, and finely granular nuclear chromatin
- Round, oval cells with nuclear molding
- High mitotic count, areas of necrosis
- +CK, +chromogranin, +synaptophysin

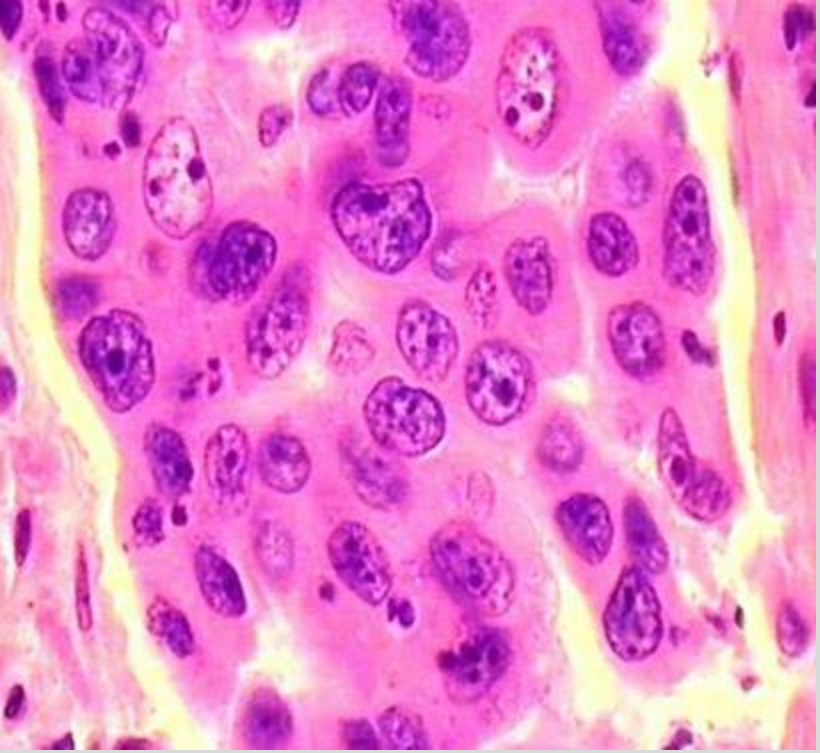
- EM shows neurosecretory granules – most likely to be associated with ectopic hormone production: **Cushing's syndrome (ACTH), SIADH - hyponatremia**

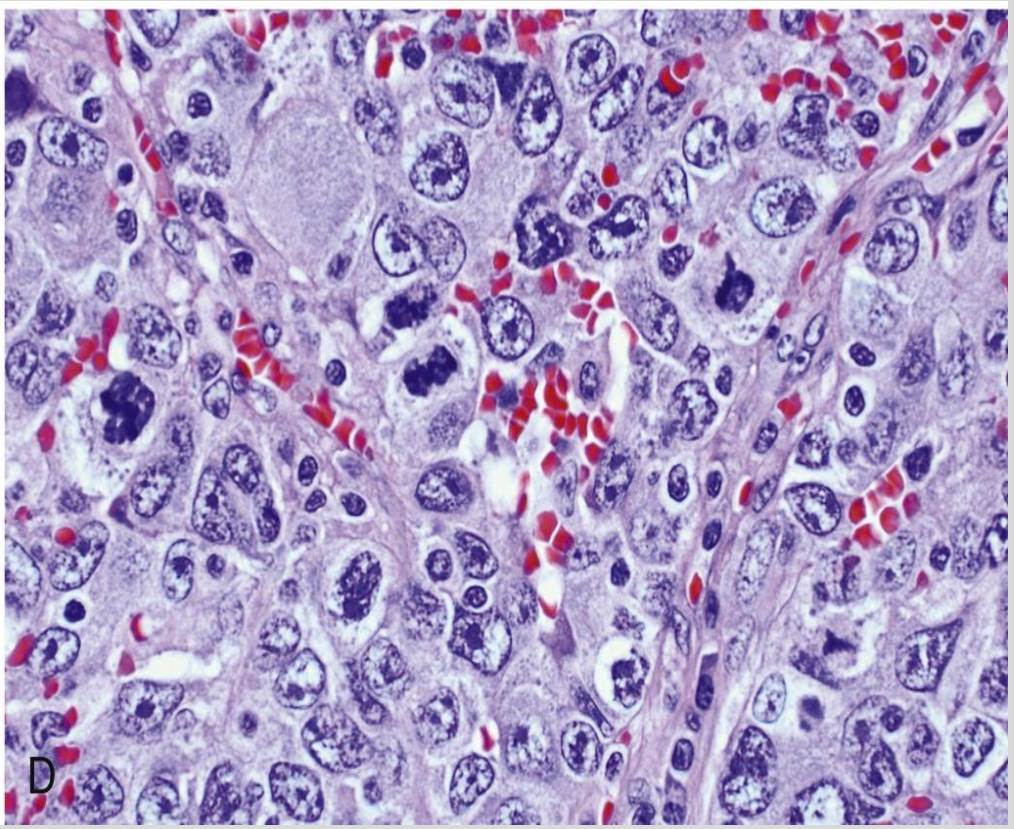
Source: Robbins and Cotran, Pathologic Basis of Disease, 11th ed, 2025, Ch. 15

Large Cell Carcinoma of lung

- Undifferentiated, malignant epithelial tumor which lacks cytology of small cell Ca and glandular or squamous differentiation; dx of exclusion
- Large nuclei, prominent nucleoli, moderate amount of cytoplasm
- May represent squamous cell or adenocarcinoma so undifferentiated that it is unrecognizable
- Do not confuse with large cell neuroendocrine carcinoma

Source: Robbins and Cotran, Pathologic Basis of Disease, 11th ed, 2025, Ch. 15





http://www.slideshare.net/dwdaugherty/lung-45773813

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Source: Robbins and Cotran, Pathologic Basis of Disease, 11th ed, 2025, Ch. 15

LOCAL effects of LUNG CANCER

MOST COMMON COUGH, HEMOPTYSIS, CHEST PAIN, WEIGHT LOSS

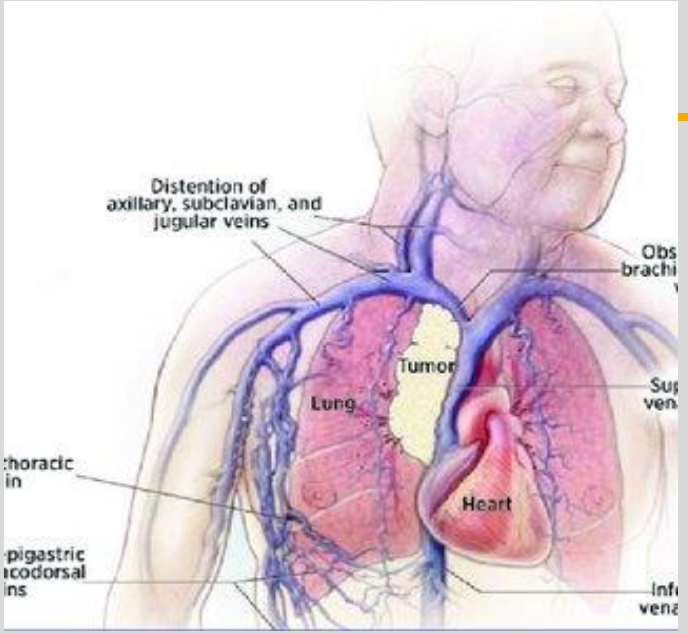
Clinical Feature:	Pathologic Basis:
Pneumonia, abscess, lobar collapse	Tumor obstruction of airway
Lipid pneumonia	Tumor obstruction; accumulation of cellular lipid in foamy macrophages
Pleural effusion	Tumor spread into pleura
Hoarseness	Recurrent laryngeal nerve invasion
Dysphagia	Esophageal invasion
Diaphragm paralysis	Phrenic nerve invasion
Rib destruction	Chest wall invasion
SVC syndrome	SVC compression by tumor
Horner syndrome	Sympathetic ganglia invasion
Pericarditis, tamponade	Pericardial involvement

Robbins, 11th ed. 2025

Secondary Pathology

- **Superior vena cava syndrome** – compression or invasion of SVC may cause **venous congestion and edema of head and arm, circulatory compromise**

- Apical (superior pulmonary sulcus) lung carcinomas known as **Pancoast tumors** may invade cervical sympathetic plexus and cause **Horner syndrome (enophthalmos, ptosis, miosis, anhidrosis)*** on same side of tumor



https://www.researchgate.net/publication/264009711_Superior_Vena_Cava_Syndrome_in_the_Cancer_Patient_A_Case_Study

*sunken in eyes, drooping lid, constriction of pupils, lack of sweating

- Focal emphysema – partial obstruction

- Atelectasis – total obstruction

- Pulmonary abscess

- Pleuritis



https://www.physio-pedia.com/Horner%27s_Syndrome

Source: Robbins and Cotran, Pathologic Basis of Disease, 10th ed, 2020, Ch. 15

SYSTEMIC effects of LUNG CANCER

(PARA-NEOPLASTIC SYNDROMES)~ 5%

ADH (hyponatremia) – ectopic production of hormone esp. in small cell carcinoma

ACTH (Cushing) - same

PTH-RP (Hyper-CA) - esp squamous cell CA

Other systemic manifestations

- Lambert-Eaton myasthenic syndrome – muscle weakness caused by autoAbs against neuronal calcium channel
- Peripheral neuropathy
- Trousseau syndrome (DVT and thromboembolism)
- Hypertrophic pulmonary osteoarthropathy – connective tissue abnormality – clubbing of fingers

Robbins and Cotran, Pathologic Basis of Disease, 11th ed. 2025

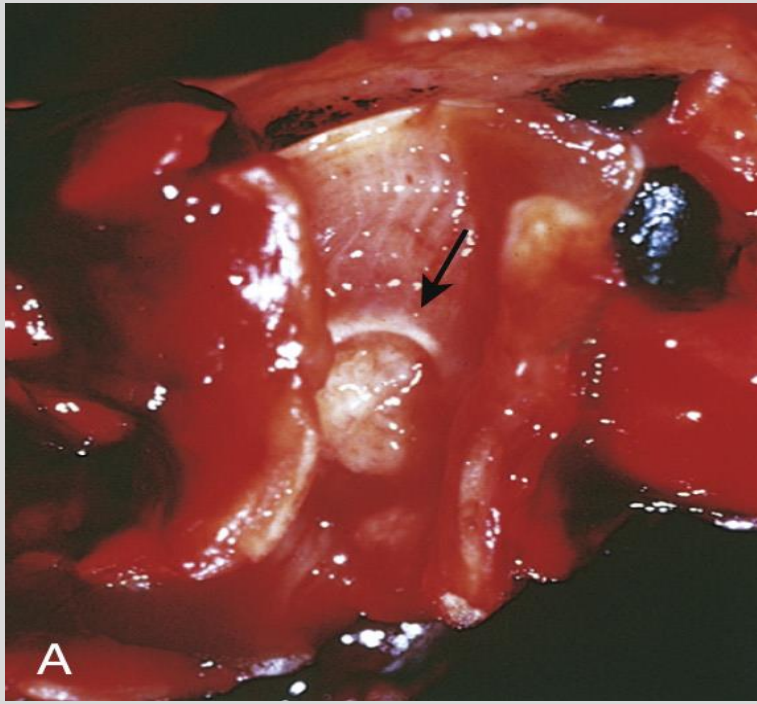
NM Staging	
Tis	Carcinoma in situ Adenocarcinoma in situ: adenocarcinoma with pure lepidic pattern, ≤3 cm Squamous cell carcinoma in situ
T1	Tumor ≤3 cm without pleural or mainstem bronchus involvement (T1mi, minimally invasive adenocarcinoma; T1a, <1 cm; T1b, 1–2 cm; T1c, 2–3 cm)
T2	Tumor 3–5 cm or involvement of mainstem bronchus but not of carina, visceral pleural involvement, or lobar atelectasis (T2a, 3–4 cm; T2b, 4–5 cm)
T3	Tumor >5–7 cm or one with involvement of parietal pleura, chest wall (including superior sulcus tumors), diaphragm, phrenic nerve, mediastinal pleura, parietal pericardium, or separate tumor nodules in the same lobe
T4	Tumor >7 cm or any tumor with invasion of mediastinum, heart, great vessels, trachea, recurrent laryngeal nerve, esophagus, vertebral body, or carina, or separate tumor nodules in a different ipsilateral lobe
N0	No metastasis to regional lymph nodes
N1	Ipsilateral intraparenchymal or peribronchial or hilar nodal involvement
N2	Metastasis to ipsilateral mediastinal or subcarinal lymph nodes
N3	Metastasis to contralateral mediastinal or hilar lymph nodes, ipsilateral or contralateral scalene, or supraclavicular lymph nodes
M0	No distant metastasis
M1	Distant metastasis (M1a, separate tumor nodule in contralateral lobe or pleural nodules or malignant pleural or pericardial effusion; M1b, single extrathoracic metastasis in a single organ; M1c, multiple extrathoracic metastases)

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Stage Grouping			
Stage 0	Tis	N0	M0
Stage IA	IA1, T1mi or T1a; IA2, T1b; IA3, T1c	N0	M0
Stage IB	T2a	N0	M0
Stage IIA	T2b	N0	M0
		N1	M0
Stage IIB	T2b	N0	M0
	T1a, T1b, T1c, T2a, T2b	N1	M0
	T3	N0	M0
Stage IIIA	T1a, T1b, T1c, T2a, T2b	N2	M0
	T3	N1	M0
	T4	N0, N1	M0
Stage IIIB	T1a, T1b, T1c, T2a, T2b	N3	M0
	T3, T4	N2	M0
Stage IIIC	T3, T4	N3	M0
Stage IVA	T any	N any	M1a, M1b
Stage IVB	T any	N any	M1c

“Carcinoid” Tumor of the Lung

- Low-grade malignant neoplasm of neuroendocrine cells (normal lung has NE cells within epithelium)
- Represent 1 – 5% of all lung tumors
- <60 y/o, ~20-40% non-smokers
- Gross appearance
  - ~3-4 cm
  - central tumor grows as a finger-like or spherical polypoid mass that often project into the mainstem bronchus
  - “Collar-button” lesion
  - Peripheral tumors are solid and nodular



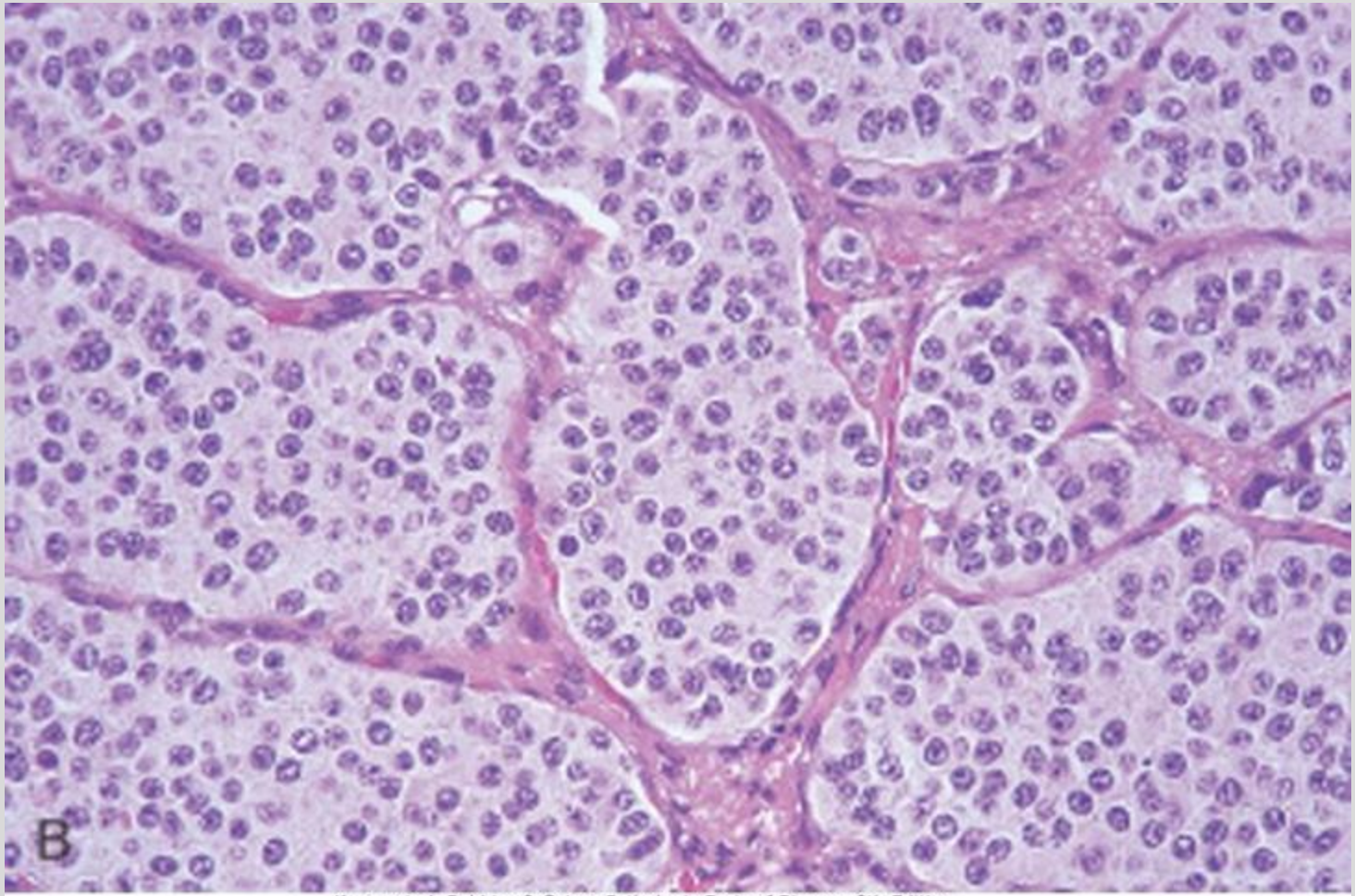
- Typical v. atypical
- about 10% produce carcinoid syndrome
  - Intermittent diarrhea, flushing, cyanosis

Source: Robbins and Cotran, Pathologic Basis of Disease, 11th ed, 2025, Ch. 15

Carcinoid

- Usually 3 to 4 cm
- Histo shows organoid, trabecular, palisading, or ribbon arrangement of cells separated by delicate fibrovascular stroma.
- Cells are regular with uniform round nuclei and moderate amount of cytoplasm.
- Typical carcinoids < 2 mitoses per 10 hpf, no necrosis – 5 yr survival rate 95%
- Atypical carcinoids between 2 and 10 mitoses per 10 hpf and/or foci of necrosis, cellular atypia, etc.- 5 yr survival rate 70%

Source: Robbins and Cotran, Pathologic Basis of Disease, 11th ed, 2025, Ch. 15



Kumar et al: Robbins & Cotran Pathologic Basis of Disease, 8th Edition.
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https://www.studyblue.com/notes/note/n/pulmonary-neoplasms-path-slides/deck/8633984

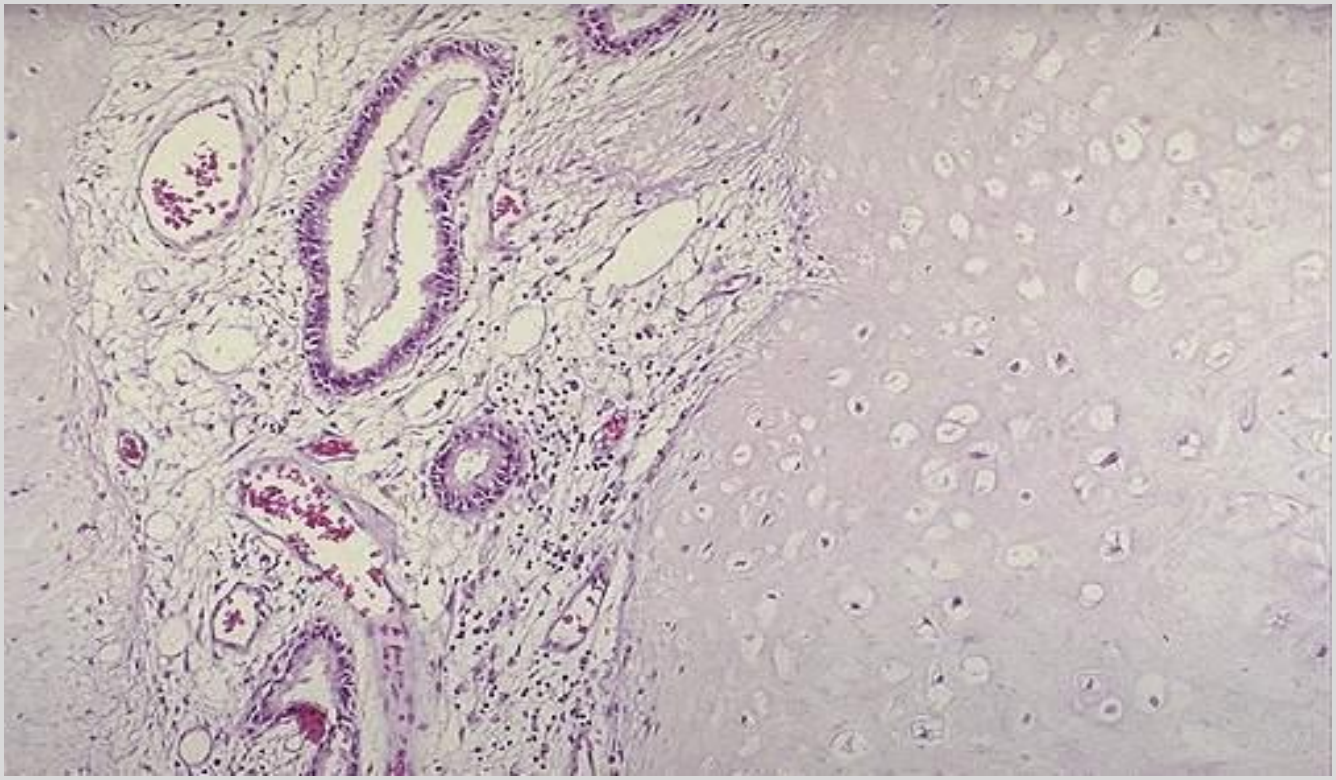
Hamartoma – made up of tissue normally found in the lung

but growing in a disorganized mass; somewhat common, benign, "coin lesion", usually peripheral, well-circumscribed, less than 3 to 4 cm. BENIGN



http://www.pathologyoutlines.com/topic/lungtumorhamartoma.html

HAMARTOMA (LUNG) – nodule of connective tissue intersected by epithelial clefts; cartilage is most common connective tissue but may also have fibrous tissue and/or fat.
Epithelial clefts lined by ciliated (or nonciliated) columnar epithelium and may represent entrapment of respiratory epithelium.



http://www.pathologyoutlines.com/topic/lungtumorhamartoma.html

Metastatic Nodules



- Tan-white nodules characteristic for metastatic carcinoma to lung (“cannonball lesions”)
- The lung is the most common site for metastatic neoplasms

https://www.studyblue.com/notes/note/n/lung-tumors--pleural-diseases/deck/9800476

Pleural diseases

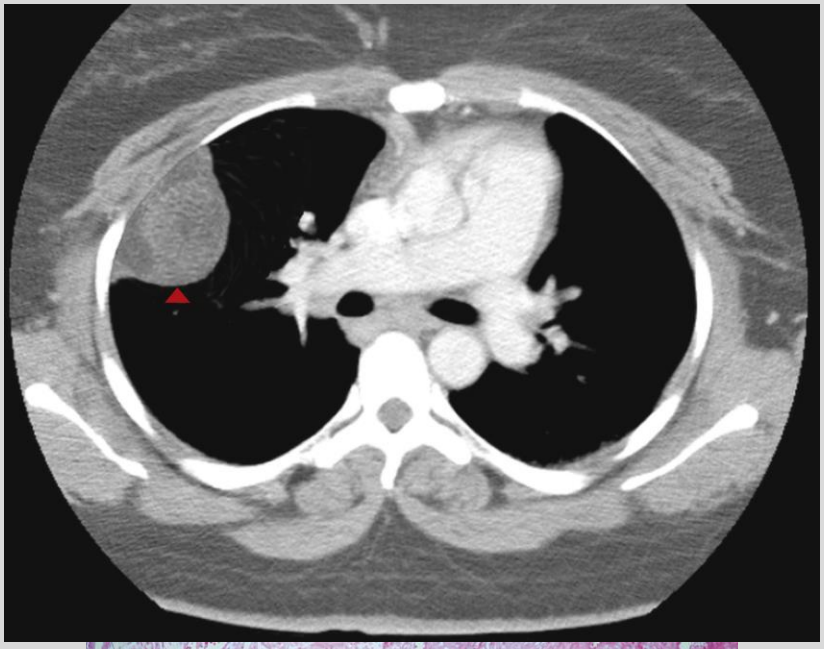
Pleural effusions – increased accumulation of pleural fluid secondary to variety of causes, malignant, benign, infection, etc.

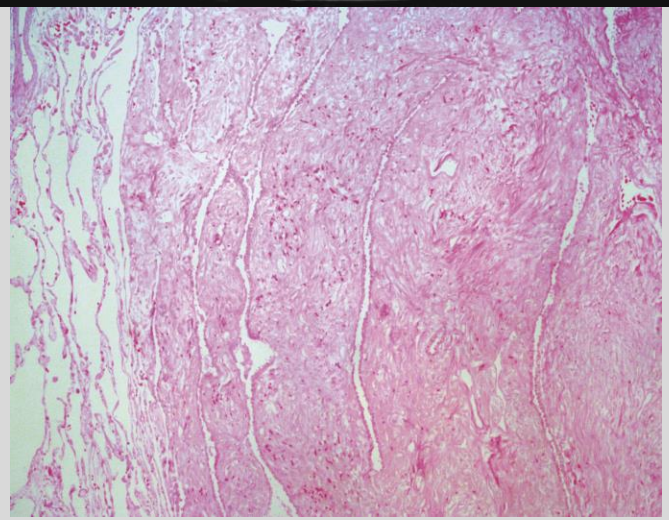
Tumors – benign (including pleural plaques), malignant, metastatic (most common)

Source: Robbins and Cotran, Pathologic Basis of Disease, 11th ed, 2025, Ch. 15

Solitary Fibrous Tumors: Benign

- Soft tissue tumor common in the pleura (less commonly in lung)
- Unrelated to asbestos exposure
- Often attached to pleura by a pedicle
- Grossly it is composed of dense fibrous tissue with occasional cysts with viscous fluid
- often attached to pleural surface by pedicle (1 – 2 cm or > 10 cm)
- NAB2::STAT6 fusion gene unique to SFT
- Rarely may be malignant
- Whorls of reticulin and collagen
- Spindle cells resembling fibroblasts
- CD34+ and STAT6+
- Negative for keratin
  - Opposite of malignant mesothelioma





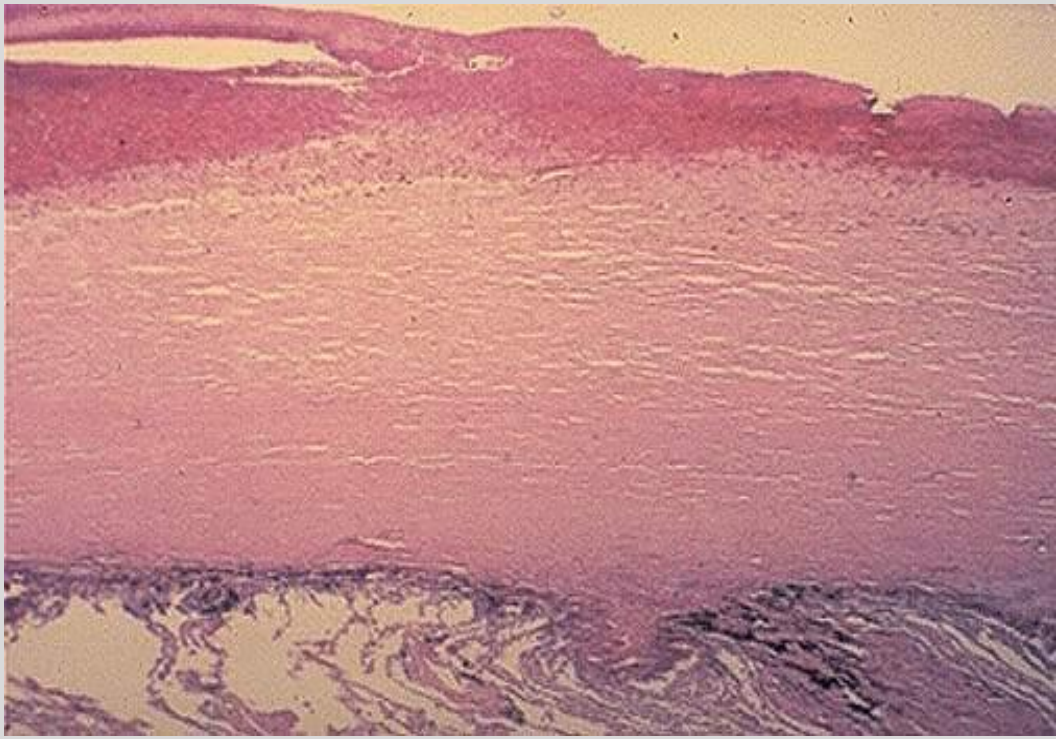
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Source: Robbins and Cotran, Pathologic Basis of Disease, 11th ed., 2025

Robbins and Cotran, Atlas of Pathology, 9th ed 2021, Ch. 5

PLEURAL PLAQUES: common manifestion of asbestos exposure. Circumscribed plaques of dense collagen, often calcified

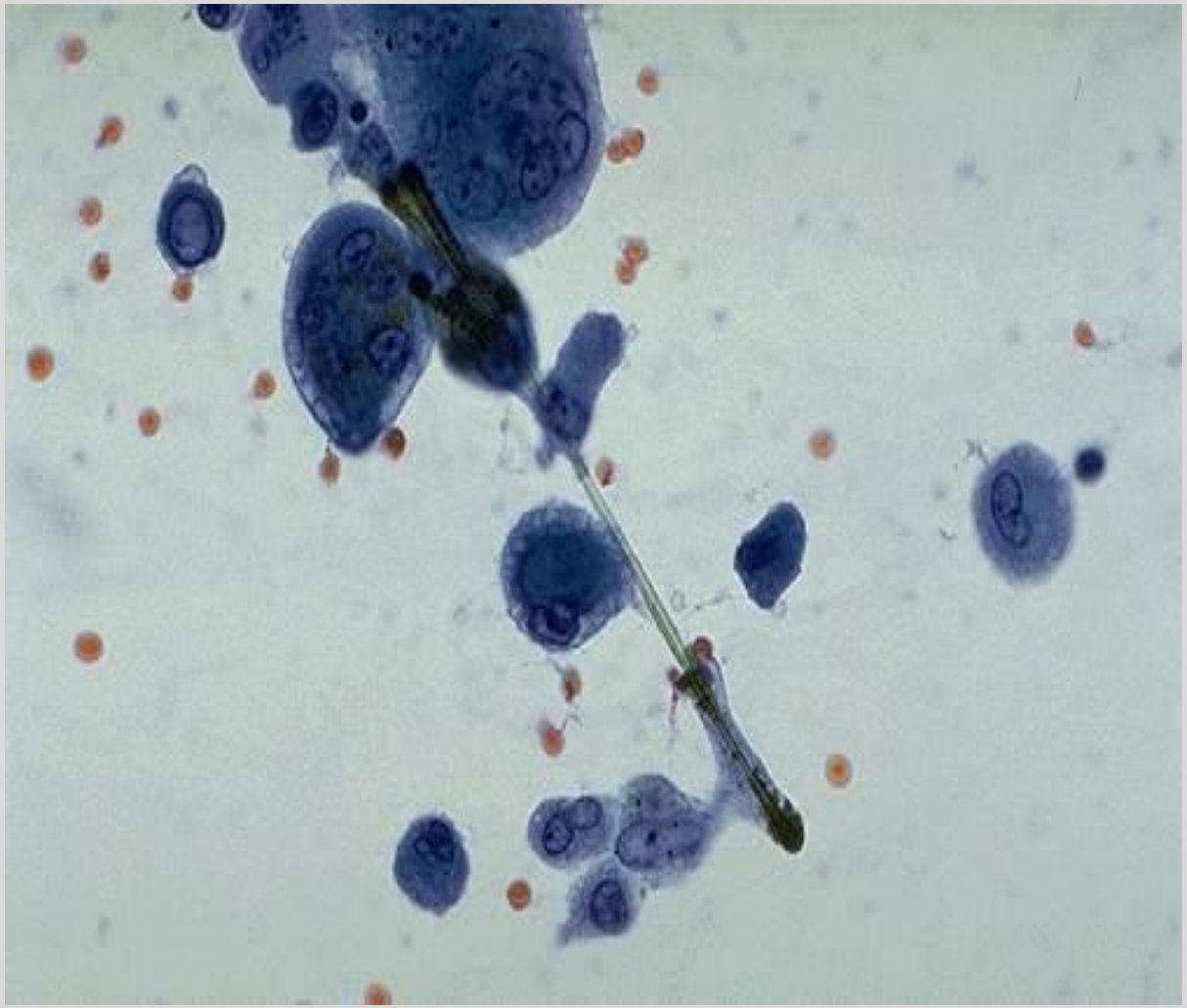




HISTOLOGY - dense layers of collagen (WebPath)

http://www.pathologyoutlines.com/topic/lungnontumorasbestosis.html

ASBESTOS BODIES



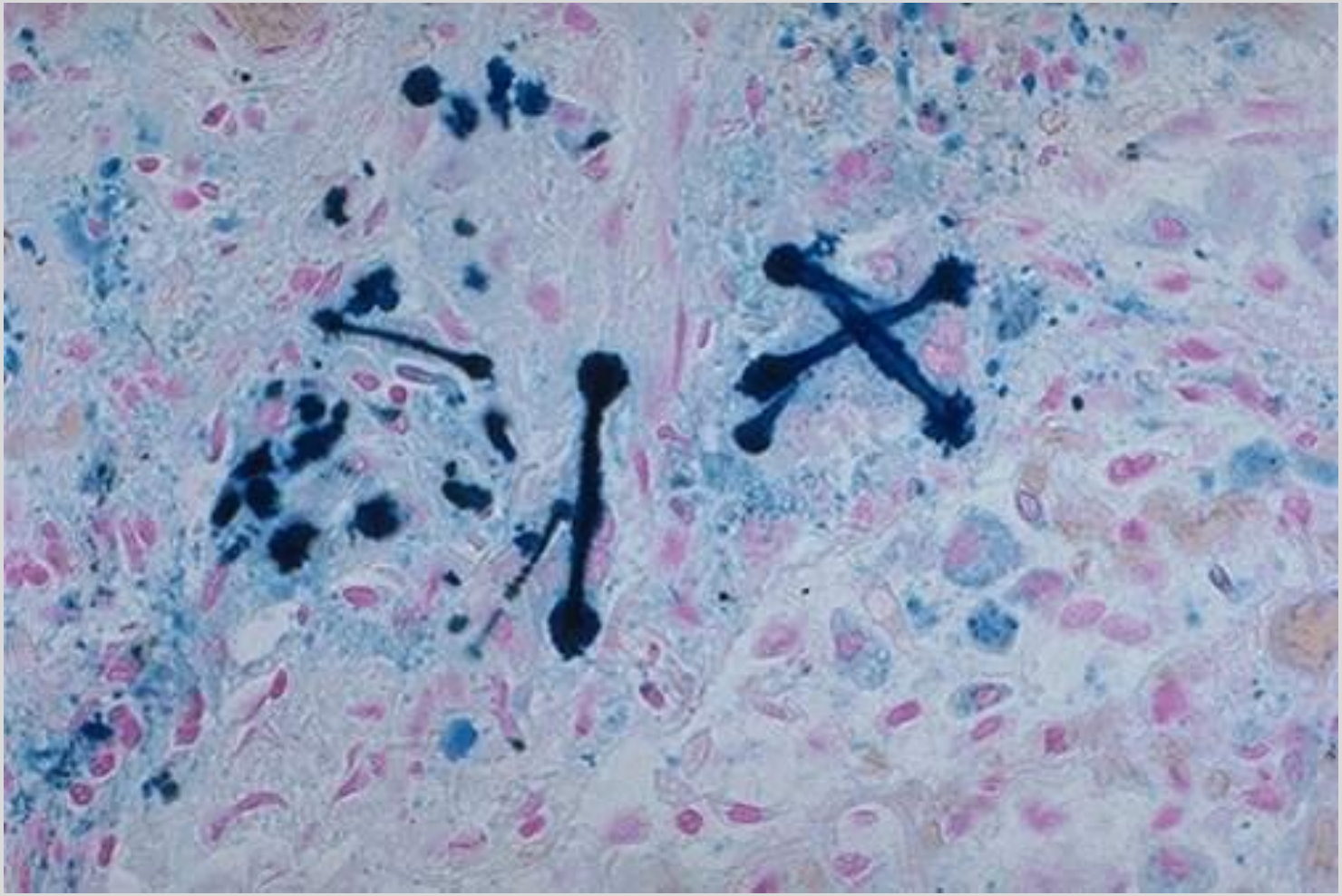
Golden brown, fusiform or beaded rods with translucent center: asbestos fibers coated with iron-containing proteinaceous material.

Macrophages try to phagocytose asbestos fibers; iron is assumed to come from phagocyte ferritin

N.B.- two forms of asbestos: serpentine and amphibole (straight and stiff); latter is less prevalent but more pathogenic

http://www.pathologyoutlines.com/topic/stainsasbestos.html

Asbestos – Ferruginous bodies
IRON STAIN (BLUE)
Other inorganic particulates may also be coated with similar iron protein complexes



http://library.med.utah.edu/WebPath/LUNGHTML/LUNG123.html

Case question

A 63 year old man who worked in a manufacturing plant for 30 years has had an 8 month history of chest discomfort, malaise, fever, night sweats, and weight loss. A chest Xray shows a pleural effusion and pleural mass encasing the lung. The patient subsequently dies of cardiorespiratory failure. Histological examination of the pleural mass at autopsy shows a biphasic pattern of epithelial and sarcomatous elements. What is the most likely diagnosis?

A. Carcinoid tumor

B. Small cell carcinoma

C. Metastatic carcinoma

D. Malignant mesothelioma

E. Large cell carcinoma

Pleural neoplasm





http://clinicalsigns-in-medicine.blogspot.com/2010/12/mesothelioma-lung.html

MALIGNANT MESOTHELIOMA

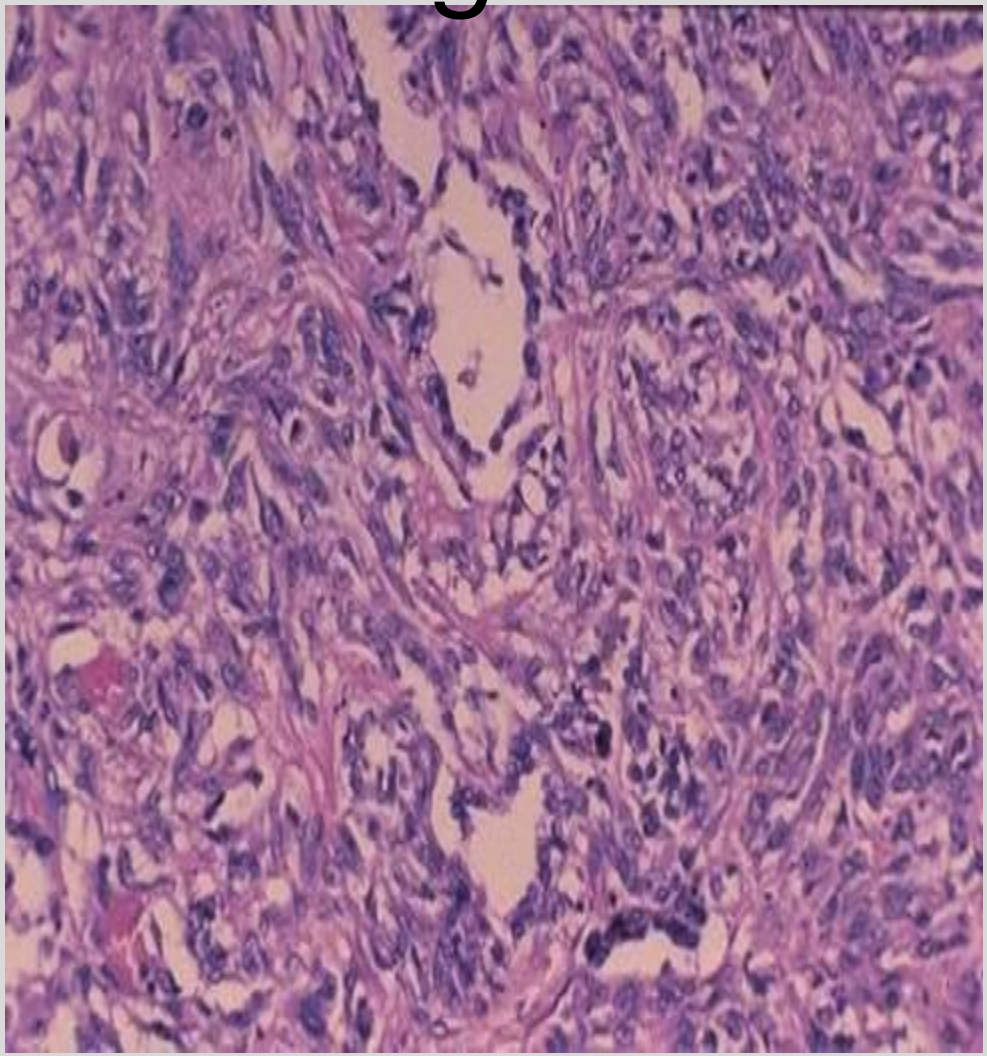


- Risk factor for mesothelioma is asbestos exposure (90% of mesotheliomas are asbestos related)
- **Latent period is 25 to 45 years.**
- **Lifetime risk of developing mesothelioma in heavily exposed asbestos individual is 7 to 10%.**

- **Asbestosis more commonly predisposes to bronchogenic carcinomas**. Smokers with a history of asbestos exposure have much greater likelihood for developing lung cancer than non-smokers. (WEBPATH)
- **Asbestos exposure, esp. if smoker, higher risk of dying from lung (bronchogenic) carcinoma**, than mesothelioma

http://library.med.utah.edu/WebPath/LUNGHTML/LUNG081.html

Malignant Mesothelioma



May arise from either visceral or parietal pleura; diffuse lesion which spreads over pleural surface and ensheaths lung with thick tumor

- **Epithelioid type (60-80%)** – cuboidal, columnar or flattened cells forming tubular or papillary structures; esp. positive for keratin proteins, calretinin, WT-1, cytokeratin 5/6 but negative for Claudin 4 (positive in adenocarcinoma)

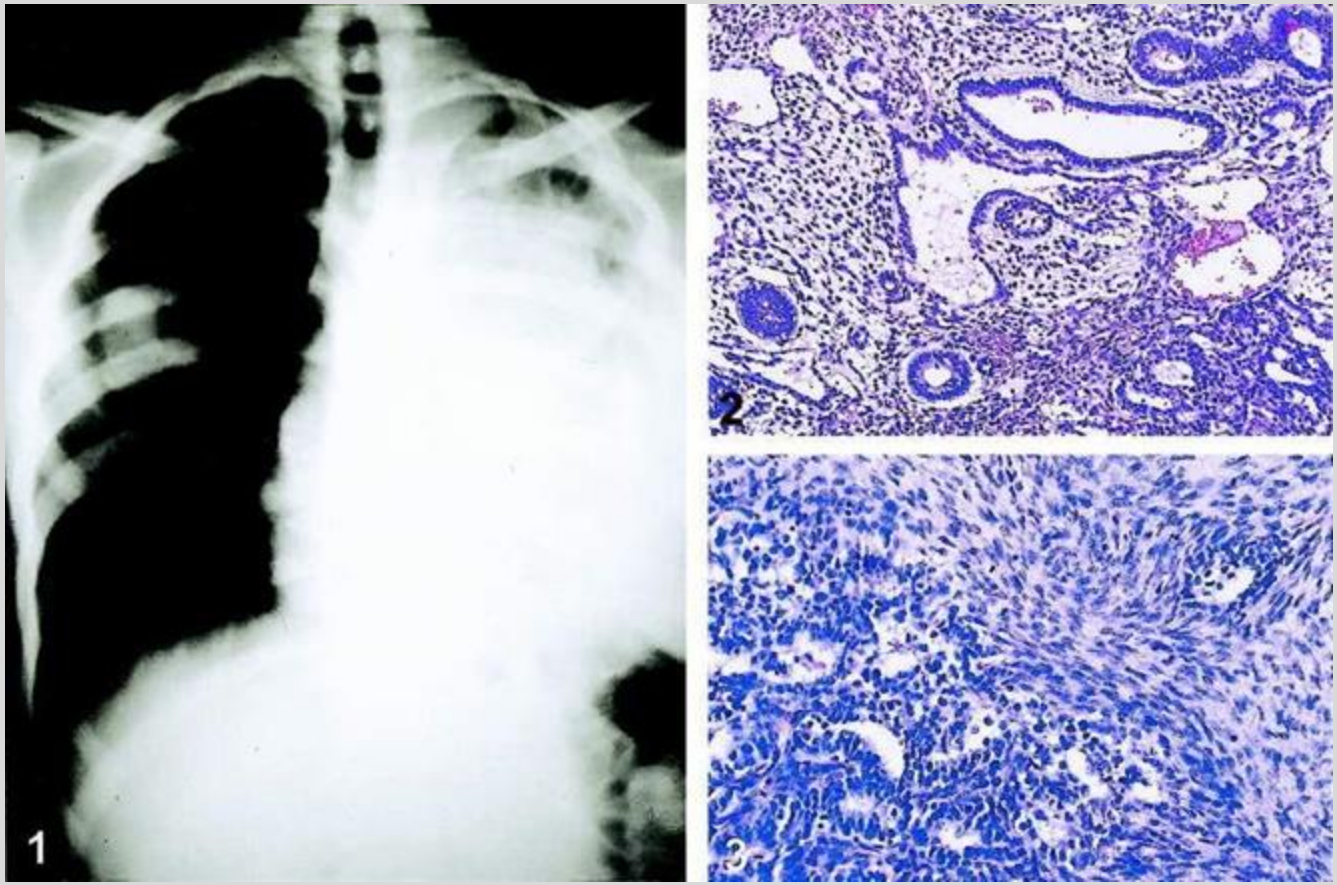
- **Sarcomatoid type** (10-12%) – mesenchymal, appearance of fibrosarcoma; lower expression of IHC markers

- **Biphasic type** (10-15%)

http://www.slideshare.net/csbrprasad/asbestosberylliosis

Source: Robbins and Cotran, Pathologic Basis of Disease, 11h ed, 2025, Ch. 15

Mesothelioma: BIPHASIC PATTERN (10 – 15%)– mixture of two types of cells; mesothelial cells can develop into either epithelium-like lining cells (epithelioid type) or mesenchymal stromal cells (sarcomatoid type). Mixed type also exists. Special stains often required for dx.



https://lh3.googleusercontent.com/kq9J46d2qNaHgrtZ-1k5SYzNGSaPAt3ySRfvdNWojcuAB_iZIIkgqfbiN57BAMujNjpR6A=s124

Malignant mesothelioma

- 90% asbestos related
- Lifetime risk in heavily exposed individuals 7 – 10%
- 80% have homozygous deletion of chromosome 9p leads to loss of tumor suppressor gene CDKN2A; also mutations in NF2, BAP1
- Epithelioid mesothelioma – 60%
- Sarcomatoid mesothelioma – 20%
- Mixed – 20%
- + calretinin, WT-1, cytokeratin 5/6, see previous slide
- 50% of patients die within 12 months of diagnosis
- Not just pleural – can arise in peritoneum and other sites

Source: Robbins and Cotran, Pathologic Basis of Disease, 11th ed, 2025, Ch. 15

THE FOLLOWING 8 QUESTIONS COME FROM ACADEMIC SCHOLAR LOGAN MEUER CLASS OF 2024, ARKANSAS CAMPUS

Case Question 1

A 59 y/o female presents with a 6-month history of mild chest pain, cough, and low-grade fever. She has no significant PMH or family history. She does not smoke. Chest radiograph reveals a small, peripheral mass in the left upper lobe. What is the most likely diagnosis?

A. Small cell carcinoma
B. Large cell carcinoma
C. Adenocarcinoma
D. Squamous cell carcinoma
E. Mesothelioma

Case Question 2

What histological features will most likely be present in this patient's tumor?

A. Pleomorphic cells with no evidence of squamous or glandular differentiation
B. Keratinization with intracellular bridging
C. Small basophilic cells with necrotic areas
D. High degree of glandular differentiation with the presence of mucin

Case Question 3

A 65 y/o male presents with a 3-month history of persistent cough, chronic fatigue and weakness. He tells you, “I can barely make it up the stairs without stopping to catch my breath”. He has smoked about two packs per day for the last 30 years and consumes 1-2 alcoholic beverages per day. Chest x-ray reveals a centrally located mass near the right mainstem bronchus. What is the most likely diagnosis?

A. Small cell carcinoma
B. Large cell carcinoma
C. Adenocarcinoma
D. Squamous cell carcinoma
E. Mesothelioma

Case Question 4

In addition to the chest x-ray, a routine CBC and BMP are ordered with the following results. What is the most likely cause of the abnormal results?

CBC	Normal Range
WBC: 5.2	4.0-10
RBC: 5.0	4.5-6.3
Hgb: 15.0	14-18
Hct: 45.0	39-52
MCV: 85.1	77-97
MCH: 30	26-32
MCHC: 34	32-36
Platelets: 225	150-450

BMP	Normal Range
Na: 142	136-144
K: 3.8	3.3-5.1
Cl: 107	98-108
CO2: 27	20-32
BUN: 10	7-22
Cr: 0.8	0.7-1.5
Glucose: 95	70-100
Ca: 15.3	8.9-10.3

A. Increased PTHrP
B. Increased ACTH
C. Increased Anti-Diuretic Hormone
D. Decreased Fluid Intake
E. Anemia

Case Question 5

A 54 y/o female presents with an 8-month history of chronic fatigue and mild depression. She has gained a significant amount of weight since her previous visit one year prior with her BMI increasing from 23 to 31. She has made no significant changes to her diet or lifestyle. PE reveals excessive hair growth on the face and neck as well as truncal obesity. CT scan of the thorax demonstrates a hilar mass. What histologic features will most likely be present in this patient's tumor?

A. Pleomorphic cells without squamous or glandular differentiation
B. Small, round basophilic cells with granular chromatin and high level of mitoses
C. Cells with uniform round nuclei and moderate cytoplasm
D. Glandular differentiation and mucin production
E. Squamous dysplasia with areas of focal necrosis

Case Question 6

If genetic testing were to be carried out on this tumor, which mutation(s) would most likely be present?

A. p53 and RB
B. EGFR and KRAS
C. FGFR-1 and CDKN2
D. ROS-1
E. BRAF and PIK3

Case Question 7

A 50 y/o male presents with complaints of difficulty breathing that has persisted for about a week. He has no significant PMH or family history. PE reveals wheezing primarily on the right side close to the midline. Upon further questioning, he admits to having several watery stools over the last several days as well. He is otherwise in good health and has never smoked. CT scan of the chest reveals a 3 cm mass that projects into the right mainstem bronchus. What is the most likely diagnosis?

A. Mesothelioma
B. Large Cell Carcinoma
C. Carcinoid Tumor
D. Small Cell Carcinoma
E. Adenocarcinoma

Case Question 8

A 75 y/o retiree is watching television when he sees a commercial for a class action lawsuit for individuals exposed to asbestos who later developed mesothelioma. He believes he may be entitled to compensation and comes to your office to confirm that he has mesothelioma. If this patient were to be diagnosed with mesothelioma, what histological features would most likely be present in his tumor?

A. Mesenchymal type cells resembling a fibrosarcoma
B. Flattened cells forming tubular or papillary structures
C. Epithelial and mesenchymal tissue with no organizing pattern
D. Dense fibrous tissue with occasional cysts
E. Nodules of connective tissue interspersed with epithelial clefts

ANSWERS TO LAST 8 CASE QUESTIONS

- 1. C. Adenocarcinoma
- 2. D. High degree of glandular differentiation with the presence of mucin
- 3. D. Squamous cell carcinoma
- 4. A. Increased PTHrP: Elevated calcium on BMP
- 5. B. Cushing's syndrome secondary to small cell carcinoma of lung
- 6. A. p53 and RB
- 7. C. Carcinoid syndrome secondary to carcinoid tumor
- 8. B. Epithelioid type mesothelioma (60-80%)

Summary Slide

- Lung cancer- Classification and overview
- Adenocarcinoma - Peripheral, mucin production, TTF1 positive
  - Females, non-smokers
  - Atypical adenomatous hyperplasia (precursor lesion)
  - Adenocarcinoma-in-situ (formerly BAC), mucinous and non-mucinous
- Squamous cell carcinoma - Central, keratin, Correlated with smoking
  - p53 mutations
  - Hypercalcemia
- Small cell carcinoma - Very aggressive, highly malignant, highest degree of correlation with smoking
  - Neuroendocrine
  - P53 and RB tumor suppressor gene mutations
  - Small cells, high mitotic rate
  - ACTH-Cushing syndrome, ADH- SIADH leading to hyponatremia
- Large cell carcinoma – undifferentiated
- Carcinoid – low grade neuroendocrine neoplasm
- Hamartoma
- Metastatic cancer to lungs
- Pleural lesions – Pleural plaques, asbestos correlation, malignant mesothelioma (associated with asbestos exposure)

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**"If smoking relaxes you, then don't quit.
Being dead is very relaxing."**

Lecture Feedback Form:

https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R